

Raidium-Rapport-TOUSNAKHOFF-DABERT

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1 Introduction

With advances in computer vision over the past 10 years, machine learning is increasingly being used in the medical field. Healthcare professionals use artificial intelligence to help them in their diagnosis or in the study of different medical documents or images. The CT-Scan is a machine that scans cuts of different parts of the human body. It is often used to detect early-stage lung cancer, colon cancer, or even heart disease.

By scanning different cuts of a body, it can be used to detect tumors or abnormal masses in the body of a patient.

2 Problem statement

The challenge proposed by Raidium consists of conceiving a model that can segment objects on one cut of a CT-Scan, highlighting the different organs, as well as potential tumors in a patient's body.

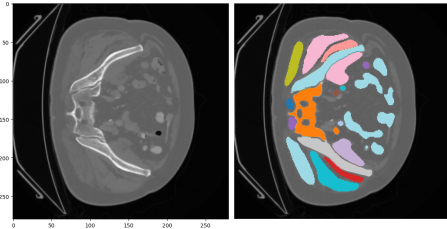


Figure 1: Example of the segmentation of a CT-Scan sample

To do so, we were provided a complete image dataset of 2000 images: 800 were partially segmented whereas the rest were not segmented. The segmen-

tation was given by another file consisting of N lines and 65536 columns. Each column corresponded to a pixel in the flattened images, and each line corresponded to one image. The different classes of segmentation of each pixel were informed in that file. In addition to that, the segmentation of the training samples is not exhaustive, which can perturb the training of models that leverage data.

To evaluate the performance of a given model, the evaluation metric that was chosen is the DICE score. For X the ground truth segmentation of a particular organ and Y the segmentation computed through our model, the DICE score is computed as follows :

$$DICE = \frac{2|X \cap Y|}{|X| + |Y|}$$

3 Segment Anything Model (SAM)

The dataset being only partially segmented with a lot of nonsegmented images, training a model from scratch would have been complicated. To overcome this issue, we choose to use a segmentation model trained on an enormous corpus of images : the Segment Anything model from Meta AI[2].

SAM finetuning

Despite its extensive training in a vast and diverse dataset, the Segment Anything Model (SAM) has not been specifically trained on medical images, particularly CT scans. The performance of SAM relies heavily on the data distribution it has seen during training. However, medical imaging, and especially CT scans, have highly specialized structures that can

differ significantly from natural images. As a result, SAM struggles to produce accurate segmentation in this context, which motivates the use of fine-tuning on domain-specific data to improve its performance in medical applications.

To finetune SAM, we first separated the annotated images in a training and testing set. We then created binary masks for each image. Using all the annotated images and their corresponding masks, we created a training loop. At each epoch, the model was trained using a combination of binary cross entropy loss between the predicted mask and the ground truth, and a score loss based on intersection over union.

After 2000 training steps, the segmentation results are much better:

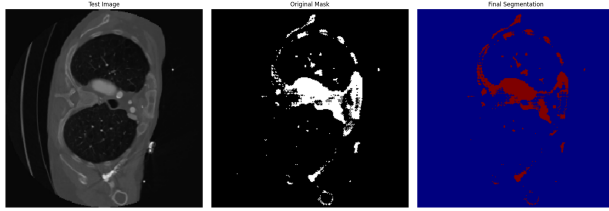


Figure 2: Results of SAM segmentation after the fine-tuning.

4 Using Yolo to prompt SAM efficiently

In this project, we used YOLO11 to generate bounding boxes that served as prompts for SAM-2. YOLO (You Only Look Once) [3] is a well-known object detection model that works by dividing an image into a grid and predicting bounding boxes and class probabilities simultaneously. Unlike models that first generate region proposals and then classify them, YOLO follows a single-stage approach, making it faster and more efficient.

To train YOLO11, we used a dataset containing annotated images with objects of different types and sizes. The training process focused on optimizing the model using a combination of cross-entropy loss for

classification and Intersection-over-Union (IoU) loss for localization. To improve generalization, we applied data augmentation techniques such as random cropping, scaling, and color jittering.

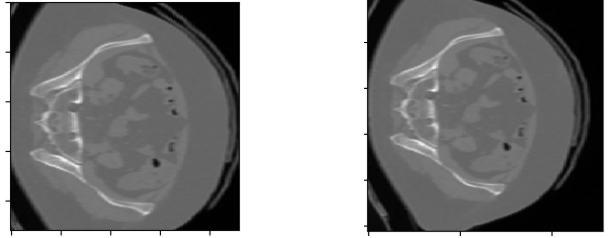


Figure 3: Image sample and its augmented version

For model training, we relied on the Ultralytics interface, which provides a direct way to train the model and monitor validation metrics such as classification loss and box loss. To use this interface, we needed to provide a dataset containing images and labels corresponding to the correct bounding boxes. We explored two different ways of labeling the bounding boxes: - Bounding boxes that enclose all pixels with the same value in the masking. - Bounding boxes that enclose all connected components of the map of pixels with the same value in the masking.

The first approach appears more suitable for classification, as it provides more features to differentiate between different organs. However, the second approach allows for drawing more precise boxes, even though identifying the specific class they belong to becomes more challenging.

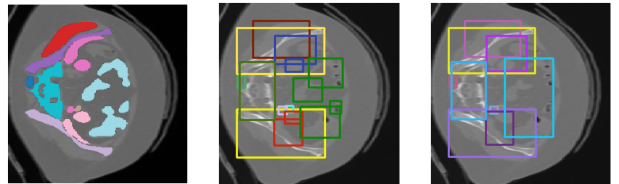


Figure 4: Different bounding box labeling strategies.

We also fine-tuned the model by adjusting hyperparameters such as the confidence threshold for object detection and applying focal loss to handle class imbalances. Additionally, we used non-maximum

suppression (NMS) to refine the final detections and ensure that the predicted bounding boxes were as accurate as possible.

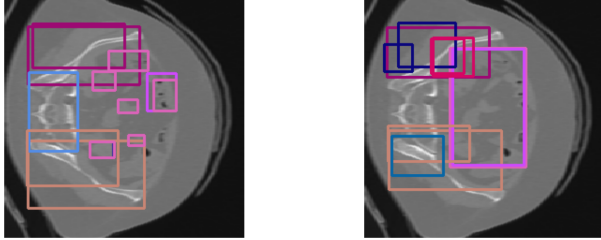


Figure 5: Inference of YOLO on a test sample.

After generating the bounding boxes with YOLO11, we fed them into SAM, which used them to produce detailed segmentation masks. The quality of these segmentations depended on the accuracy of YOLO11’s detections. Our results showed that well-calibrated bounding boxes led to significantly better segmentation outputs, especially in images with overlapping objects or cluttered backgrounds. With that new pipeline, we finetuned SAM with bounding boxes delimiting the labeled mask as prompt. This resulted in better capacity of SAM to segment the scans.

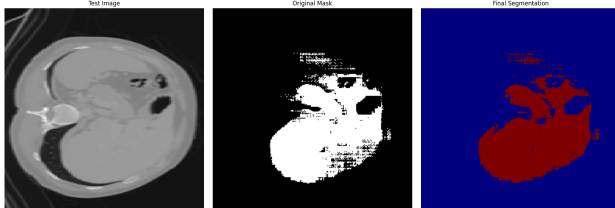


Figure 6: SAM segmentation with bounding boxes as prompt

5 Semi-supervised Learning

In addition to fine-tuning pretrained models, we also explored semi-supervised learning as an alternative approach to enhance model performance. This method enables the model to learn from both labeled and unlabeled data, improving generalization

and robustness. For this purpose, we employed the FreeMatch algorithm, as described in [4].

The FreeMatch algorithm constructs training batches that include samples from both the labeled dataset and the unlabeled dataset. For labeled samples, the model follows the standard supervised learning process by making predictions and computing losses. For unlabeled samples, the model first generates predictions and evaluates their confidence. If the confidence surpasses a predefined threshold, the predicted labels are temporarily assigned to the samples. The model then makes another prediction on an augmented version of the image, encouraging consistency between different views of the same sample.

A key feature of FreeMatch is its adaptive thresholding mechanism, which dynamically adjusts the confidence threshold based on the model’s predictions. This ensures a balanced integration of labeled and unlabeled data, gradually increasing the reliance on pseudo-labels as training progresses. By leveraging information from both labeled and unlabeled data, this approach enhances the model’s ability to generalize to new instances, leading to improved overall performance.

6 Results

In this section, we analyze the performance of our approach both qualitatively and quantitatively. The qualitative evaluation focuses on the visual quality of the segmentation masks, while the quantitative evaluation provides numerical performance metrics such as Intersection over Union (IoU) and the Dice coefficient.

Qualitative Analysis

To assess the quality of the segmentation masks produced by SAM when prompted with YOLO-generated bounding boxes, we visually inspect the predicted masks and compare them with the ground truth annotations. Representative examples from the dataset are shown in Figure 7.

We observe that the segmentation quality is highly dependent on the precision of the bounding box

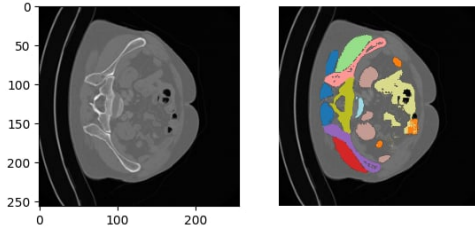


Figure 7: Example segmentation outputs using YOLO-generated prompts for SAM.

proposals. When YOLO accurately detects object boundaries, SAM is able to generate precise segmentations. However, in cases where YOLO’s bounding boxes are too coarse or misaligned, the segmentation masks tend to suffer, particularly in regions with fine details or overlapping structures.

Quantitative Analysis

To quantitatively evaluate the segmentation performance, we compute standard metrics such as the Intersection over Union (IoU) for the segmentation of SAM and the Dice similarity coefficient (DSC). These metrics measure the overlap between the predicted masks and the ground truth annotations.

Method	Mean IoU	Mean Dice Score
SAM2	0.6	0.16
YOLO+SAM2	0.75	0.27
FreeMatch*	0.76	0.30

Table 1: Quantitative evaluation of segmentation performance.

(*Same pipeline as above with semi supervised fine-tuning)

From Table 1. The improvement highlights the advantage of using high-quality bounding box prompts to guide the segmentation process. However, we note that performance degrades on objects with complex structures, suggesting potential areas for further refinement, such as incorporating additional refinement modules or post-processing techniques. Moreover, even though the detection with Yolo permits to de-

tect a great number of objects, it fails at classifying them well.

7 Discussion and Analysis of Results

The evaluation of our approach on the private leaderboard yielded a Dice score of 0.3041, corresponding to rank 7. While our qualitative results appear satisfying, the current limitations in post-processing and a suboptimal exploitation of the available annotations during training have likely prevented us from achieving a higher ranking.

Strengths and Limitations

The integration of YOLO for generating bounding box prompts significantly enhanced SAM’s segmentation performance by guiding it toward regions of interest. This method allowed for improved segmentation quality when object boundaries were accurately detected. However, our training process did not fully leverage all the available annotations, resulting in an underutilization of potentially valuable training information.

Opportunities for Improvement

Future work should focus on:

- **Enhanced Post-Processing:** Incorporating advanced post-processing techniques could further refine segmentation outputs, especially in regions with overlapping or ambiguous structures.
- **Better Annotation Utilization:** Developing strategies to more effectively use the annotations during training—such as dynamic weighting in the loss function or improved bounding box extraction methods—could lead to significant performance improvements.

8 Possible further improvements : Self-Supervised Finetuning

One possible improvement, that leverages the non-annotated data, and that can be used to further enhance the robustness and the comprehension of the model on CT-Scan data is self-supervised learning. Inspired by the work at Meta AI on self-supervised transformers[1], our idea, that we did not have time to implement, was as follows :

- For each image in the dataset, generate different augmentations of this image, by cropping, rotating, blurring and/or scaling.
- At training time, use two models : a teacher and a student. These are both sam models but with different parameters.
- At each epoch, we pass through the teacher and the student two different augmentations of the same image, and we gather the segmentation results.
- We then apply a loss that compares both segmentations, while making sure that the gradients are only propagated through the student model. The teacher model is updated through exponential moving average of the instances of the student model.

We note that it would also be possible to realize this specific training with the encoder part of the SAM model, comparing the latent representations of the student and the teacher.

As shown by Meta, this kind of approach helps the model to enhance its robustness and also his comprehension of different images. As SAM wasn't necessarily trained on medical data, this could be a fruitful approach to enhance its results and its segmentations in that field.

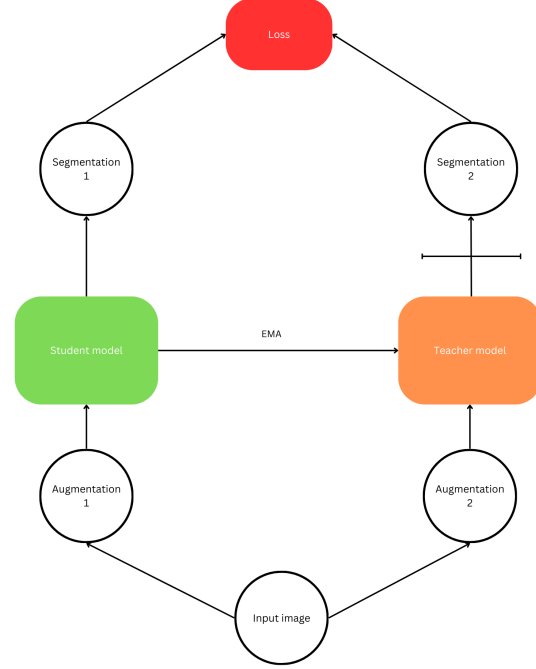


Figure 8: Self Supervised Learning algorithm

9 Conclusion

In this work, we addressed the challenge of segmenting CT-Scan images by integrating a fine-tuned version of the Segment Anything Model (SAM) with YOLO-generated bounding box prompts. Our approach demonstrated that guiding SAM with accurate bounding boxes significantly improves segmentation quality. The evaluation on the private leaderboard yielded a Dice score of 0.3041, corresponding to rank 7. Although our qualitative results are promising, the lack of advanced post-processing and a more effective use of the available annotations limited further improvements.

Future work will focus on incorporating refined post-processing techniques and optimizing the anno-

tation integration during training, as well as exploring self-supervised fine-tuning strategies. These enhancements are expected to further improve the robustness and accuracy of the segmentation in medical images.

References

- [1] Mathilde Caron et al. “Emerging Properties in Self-Supervised Vision Transformers”. In: *Proceedings of the International Conference on Computer Vision (ICCV)* (2021).
- [2] Alexander Kirillov et al. “Segment Anything”. In: *arXiv preprint arXiv:2304.02643* (2023).
- [3] Joseph Redmon et al. “You Only Look Once: Unified, Real-Time Object Detection”. In: *CVPR* (2016).
- [4] Yidong Wang et al. “FreeMatch: Self-adaptive Thresholding for Semi-supervised Learning”. In: *ICLR* (2023).